

CRISTIANO ULONDU MENDES

🌐 cristiano-ulondu-mendes.github.io/personal-website

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🏳️ nationality: Bissau-guinean

EDUCATION

PhD in Signal and Image Processing *Palaiseau, France*
Thesis: Deep learning for Synthetic Aperture Radar (SAR) data processing and analysis *2023 -- present*
Topics: (polarimetric) (interferometric)-SAR despeckling, tomographic SAR reconstruction,
LTCI, Télécom Paris, Institut Polytechnique de Paris

Master's Degree in Mathematics, Data, and Learning *Paris, France*
Track: Mathematics, Modeling, Learning *2021 - 2022*
Paris Cité University
with highest honour

Bachelor's Degree in Fundamental and Applied Mathematics *Paris, France*
2020 - 2021
Paris Cité University
with high honours

SKILLS

Programming languages:	Python, MATLAB, $\text{\LaTeX}$
Frameworks & Libraries:	PyTorch, NumPy, SciPy, OpenCV
Research areas:	Deep learning, inverse problems, sparse reconstruction, variational methods, SAR imaging, polarimetry, tomography
Tools:	Git, GitLab, HPC/cluster computing
Operating systems:	Linux, Windows

EXPERIENCE

- Master 1 semester project *01/02/21 - 17/05/21*
Paris Cité University, France
Subject: State-of-the-art study of inverse problem-solving methods for medical imaging
Tasks performed :
 - bibliographic study of inversion methods for tomographic imaging
 - implementation of variational methods and comparison on simulated data (Sheep-Logan phantom)Developed skills:
 - literature review and critical analysis of scientific papers
 - numerical implementation and evaluation of mathematical methods
 - understanding of inverse problems and variational approaches
- End of studies internship *01/03/22 - 31/08/22*
Laboratoire Traitement et Communication de l'Information (LTCI), Télécom Paris, France
Subject: Exploration of variational approaches for multi-temporal despeckling of polarimetric Synthetic Aperture Radar (SAR) images.

Tasks performed :

- bibliographic study on SAR imaging
- porting the Matlab code of a single-date SAR image denoiser (MuLoG) to Python
- proposals for an extension to polarimetric SAR images of a multi-temporal denoising scheme initially developed for single-channel SAR images (RABASAR)
- proposal of solutions to account for noise spatial correlation in the despeckling of real SAR images.

Developed skills:

- scientific programming in Python and MATLAB
- adaptation and implementation of methods from scientific literature
- image processing and statistical noise modeling
- autonomous research and problem solving

- Research engineer 15/10/22 - 31/08/23
Laboratoire Traitement et Communication de l'Information (LTCI), Télécom Paris, France

Subject: Extension of the end-of-studies internship

Tasks performed :

- writing of my first journal paper based on the internship results
- answering reviewers' comments through clarifications and additional experiments

Developed skills:

- scientific writing and publication
- experimental design and quantitative evaluation
- critical analysis and interpretation of research results
- response to peer-review comments

PUBLICATIONS

Journal Articles

- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Fast tomographic SAR inversion by unrolling a greedy algorithm for sparse reconstruction,`` *submitted to IEEE Transactions on Geoscience and Remote Sensing*, June 2026. [Preprint: hal-05609504]
- C. Ulundu Mendes, E. Dalsasso, L. Denis, F. Tupin, ``POLARSAR2POLARSAR: A semi-supervised despeckling algorithm for polarimetric SAR images,`` *ISPRS Journal of Photogrammetry and Remote Sensing*, 2025. DOI: 10.1016/j.isprsjprs.2025.01.008
- C. Ulundu Mendes, L. Denis, C. Deledalle, F. Tupin, ``Robustness to spatially-correlated speckle in Plug-and-Play PolSAR despeckling,`` *IEEE Transactions on Geoscience and Remote Sensing*, 2024. DOI: 10.1109/TGRS.2024.3432180

Conferences & Workshops

- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Leveraging LiDAR datasets to improve SAR tomography: a diffusion model approach,`` *14th IAPR Workshop on Pattern Recognition in Remote Sensing*, Lyon, France, Aug 2026.
- C. Ulundu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Reconstruction 3D en tomographie radar : apprentissage profond basé sur un Matching Pursuit déroulé,`` *GRETSI 2025*, Strasbourg, France, Aug 2025.
- C. Ulundu Mendes, L. Denis, F. Tupin, ``Schéma Plug-and-Play robuste à la corrélation spatiale du speckle pour le filtrage des données SAR polarimétriques,`` *GRETSI 2023*, Grenoble, France, Aug 2023.